

Hierarchical Semantic Perceptual Listener Head Video Generation: A High-performance Pipeline

Zhigang Chang*

Shanghai Jiao Tong University
Du Xiaoman Financial
Shanghai-Beijing, China
changzig@sjtu.edu.cn

Qing Yang

Du Xiaoman Financial
Beijing, China
yangqing@duxiaoman.com

Weitai Hu

Du Xiaoman Financial
Beijing, China
huweitai@duxiaoman.com

Shibao Zheng

Shanghai Jiao Tong University
Shanghai, China
sbzh@sjtu.edu.cn

ABSTRACT

In dyadic speaker-listener interactions, the listener's head reactions, together with the speaker's head movements, form an important non-verbal semantic expression. The listener head generation task aims to synthesize the responsive listener's head videos based on audios of the speaker and reference images of the listener. Compared to the talking-head generation, it is more challenging to capture the correlation cues from the speaker's audio and visual information. Following the ViCo baseline scheme, we propose a high-performance solution by enhancing the hierarchical semantic extraction capability of the audio encoder module and improving the decoder part, renderer and post-processing modules. Our solution gets the first place on the official leaderboard for the track of listening head generation. This paper is a technical report of ViCo@2023 Conversational Head Generation Challenge at the ACM Multimedia 2023 conference.

CCS CONCEPTS

• Computing methodologies → Reconstruction.

KEYWORDS

Conversational Head Generation

ACM Reference Format:

Zhigang Chang, Weitai Hu, Qing Yang, and Shibao Zheng. 2023. Hierarchical Semantic Perceptual Listener Head Video Generation: A High-performance Pipeline. In *Proceedings of the 31st ACM International Conference on Multimedia (MM '23)*, October 29–November 3, 2023, Ottawa, ON, Canada. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3581783.3612869>

*Corresponding Author

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

MM '23, October 29–November 3, 2023, Ottawa, ON, Canada

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-0108-5/23/10...\$15.00

<https://doi.org/10.1145/3581783.3612869>

1 INTRODUCTION

In interactive conversational scenarios, facial expressions, head movements and micro-expressive reactions of both the speaker and the listener are capable of conveying important non-verbal semantics. Humans are able to accurately capture these non-verbal semantics during verbal communication and respond in the same way. In dyadic speaker-listener interactions, we expect machines to be similarly able to perceive the speaker's verbal and non-verbal semantics in multiple dimensions and to simulate and generate the listener's head response. This task is called "Responsive Listener Head Generation" and has great potential for applications in areas such as human-computer interaction, sentiment analysis, animation, and metaverse.

Compared to the talking-head generation [18, 22], there is much less research on the listener's head generation. This is because the mapping between speech and the speaker's facial expressions and lip movements can be learned more rigorously, while the listener's reaction to the speaker's audio and visual input is relatively ambiguous, i.e., different head reactions can be appropriate and reasonable. However, if we carefully compare the videos of the speaker and the listener in an interaction, we can still find that the listener's non-verbal semantic behaviors (such as nodding, smiling, and head shaking) are a feedback to the speaker's hierarchical multimodal semantics (such as linguistic emotions, speech rhythms, expressions, and eye contact) and involve personal behavioral styles. The question now is how to extract the correlation cues between the speaker's audio and video information and the listener's non-verbal semantic behaviors.

Inspired by [9], we adopt a Hierarchical Audio Encoder to fully exploit the semantics of the different levels. We believe that this encoder can capture both high-level speech semantics (e.g., the emotion and tone conveyed by the content) and low-level speech semantics (e.g., the rhythm and pitch of the speech) and adapt them to the listener's responsive behaviors. Similar to the ViCo baseline [20], we also use the facial reconstruction model [6] to extract the coefficients of the three-dimensional morphable model (3DMM) of the speaker [1, 12] as a visual signal encoder. A cascaded bi-directional GRU network acts as a decoder to predict the sequential outputs of the listener head coefficients. In the final rendering and post-processing phase, we employ an advanced renderer [17]

and a video restoration module [8] to improve the quality of the generated videos.

In general, this paper presents a high-performance solution to the task of generating Responsive Listener Head. Our main techniques include:

- We adopt a Hierarchical Audio Encoder to capture hierarchical speech semantics, and develop an audio feature fusion module to fuse audio features.
- We adopt a cascaded bi-directional GRU as a decoder to predict the sequential outputs. Various training techniques are applied to the limited training data.
- We adopt an improved renderer and video restoration module to significantly improve the quality of the generated videos.

2 RELATED WORK

Audio-Driven human-centric synthesis tasks [2, 5, 9, 13, 18, 21, 22] have attracted extensive research interest in recent years. The Audio-to-lip task [13] aims to generate a lip-synchronization video clip with a initial frame input. The Talking-head generation task [18, 22] tends to output a vivid talking video of the speaker with head movements from a speaker's initial frame and a clip of audio. The Audio-to-gesture task [5, 9] tries to learn mappings between audios and body languages. The commonality of the above generation tasks are that they are all based on the role of the speaker, and they all need to build the associations between speech semantics and target behaviors. Listening head generation task [10, 19, 20] is a non-verbal facial reaction prediction task. Recent facial reaction studies [11, 20] frequently represent facial motions with 3DMM coefficients (e.g., expression and pose coefficients) to better visualize the facial muscle movements. Meanwhile, speaker's speech and audio behaviours were also additionally employed as the input to deliver richer verbal and non-verbal speaker behavioural cues [11, 15, 16].

3 TASK OVERVIEW

3.1 Definition

We first briefly introduce the task definition of Responsive Listening Head Generation. Given an input video sequence $\mathcal{V}_t^s = \{v_1^s, \dots, v_t^s\}$ of the speaker head in time stamps ranging from $\{1, \dots, t\}$, and the corresponding audio signal sequence $\mathcal{A}_t^s = \{a_1, \dots, a_t\}$, the goal of this task is to generate the whole listener's head video sequence $\mathcal{V}_t^l = \{v_2^l, \dots, v_{t+1}^l\}$, where output of the next time stamp can be denoted as Eq. 1 and $\mathcal{V}_1^l$ is the initial reference frame.

$$\mathcal{V}_{t+1}^l = G\left(\mathcal{V}_t^s, \mathcal{A}_t^s, \mathcal{V}_1^l\right) \quad (1)$$

3.2 Dataset Description and Evaluation Metrics

This ViCo@2023 challenge is based on the extended ViCo dataset. The training set contains 440 pairs of speaker-listener videos with 83 listener IDs and 65 speaker IDs. The listening head test set contains 45 audio clips, speaker video clips and several listener reference images.

The quality of generated videos will be quantitative evaluated from the perspectives of visual quality, naturalness, and task-specified goals. The evaluation meaning of each indicator is as follows:

The **PSNR** (Peak Signal to Noise Ratio) and **SSIM** (Structure Similarity Index Measure) can evaluate the quality of generated images by comparing with GT images. The **CPBD** (perceptual-based no-reference objective image sharpness metric) can evaluate the sharpness of the generated images without GT images. The **FID** (Fréchet Inception Distance) evaluates the similarity of the generated images and GT in feature level. The **CSIM** is evaluated by extracting identity features predicted by ArcFace and calculate the cosine similarity. The **ExpL1** and **PoseL1** evaluate L1 distances of expression and pose coefficients from 3DMM reconstruction. The **ExpFD** and **PoseFD** evaluate Fréchet distances of expression and pose coefficients from 3DMM reconstruction.

4 METHOD

4.1 Framework

The overall pipeline is illustrated in Fig. 1 The pipeline is composed of an audio encoder, a visual encoder, a sequential decoder, a video renderer and a post-processing module. Then we will introduce each component.

4.2 Visual Encoder

Following the ViCo baseline scheme [20], 3DMM coefficients are extracted by a face reconstruction model [6] for each frame. We get 3DMM coefficients $\xi_t = (\alpha_t, \beta_t, \epsilon_t, \gamma_t, p_t)$ for each time stamp where the identity-independent features can be represented parametrically by expression and pose coefficients $\{\beta \in \mathbb{R}^{64}, p \in \mathbb{R}^6\}$.

4.3 Audio Encoder

We adopt the Hierarchical Audio Encoder E_a in HA2G [9] to extract multi-level semantic features. Transcript can be directly recognized by Automatic Speech Recognition (ASR) models from audio $\mathcal{A}_t^s$. High-level linguistic features extracted by text encoder from transcripts t_{text} as $f_t = E_{text}(t_{text})$. E_a is a ResNetSE34 borrowed from [3]. Specifically, we define the features output from ResNet Stage-2 as low-level features f_a^{low} , features output from ResNet Stage-3 as middle-level audio features f_a^{mid} , features output from ResNet Stage-4 as high-level audio features f_a^{high} . We propose to learn the association between provided transcripts and audios with contrastive learning. The training strategy is to leverage the natural synchronization between text and audio. The multi-level contrastive loss are computed by Eq. 2 to enforce the high-level audio features f_a^{high} aligned with the text features f_t .

$$L_{const} = -\log \frac{\exp\left(\text{sim}\left(f_t, f_{a+}^{high}\right) / \tau\right)}{\sum_{i=1}^K \sum_{l \in L} \exp\left(\text{sim}\left(f_t, f_{a(i)}^l\right) / \tau\right)} \quad (2)$$

where $L = \{\text{low}, \text{mid}, \text{high}\}$ and $\text{sim}(f_1, f_2) = \frac{f_1 \cdot f_2}{\|f_1\| \|f_2\|}$ is cosine similarity. f_{a+}^{high} is the high-level audio features aligned to the transcript as the positive samples. Negative samples are collected from high-level audio features of other time stamps or non-high-level semantic audio features. τ is the temperature parameter.

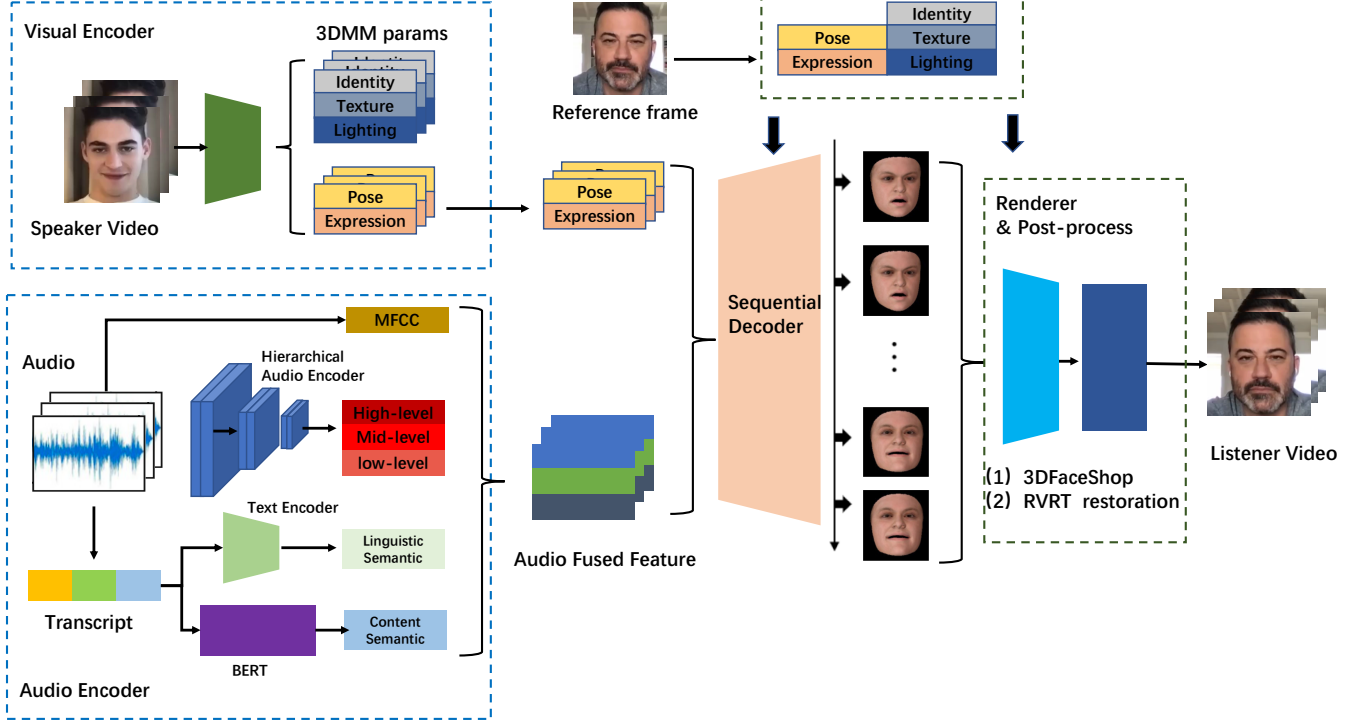


Figure 1: The Pipeline of Hierarchical Semantic Perceptual Listener Head Video Generation.

Table 1: The official final leaderboard for the track of listening head generation in ViCo@2023 challenge (results of top-3 teams are shown here).

Team	SSIM $\uparrow$	CPBD $\uparrow$	PSNR $\uparrow$	FID $\downarrow$	CSIM $\uparrow$	PoseL1 $\downarrow$	ExpL1 $\downarrow$	PoseFD $\downarrow$	ExpFD $\downarrow$	score $\downarrow$
Ours	0.647	0.163	26.757	30.605	0.669	0.070	0.139	0.009	0.936	7
Robo Space	0.612	0.186	19.064	30.982	0.676	0.078	0.149	0.024	0.604	9
ilearn	0.597	0.168	18.398	29.134	0.636	0.085	0.154	0.012	0.618	15

We also implement the pre-trained language model BERT [4] on transcripts to get the content semantics $S_T = \{s_1, \dots, s_t\}$. Mel-frequency cepstral coefficients (MFCC) features with the corresponding MFCC Delta and Delta-Delta (second-order difference) features are extracted from the original speech as well, denoted as $M_T = \{m_1, \dots, m_t\}$.

An audio feature fusion module f_{A-FFM} is designed to obtain a fusion of multiple speech features $f_a^{high}, f_a^{mid}, f_a^{low}, S_T$ and M_T . We adopt a SE-ResNet Module [7] as the audio feature fusion module to capture the channel attention. The output of the audio encoder can be denoted as follows:

$$A_t^{AE} = f_{A-FFM}(f_{at}^{high}, f_{at}^{mid}, f_{at}^{low}, S_t, M_t) \quad (3)$$

4.4 Sequential Decoder

After acquiring audio and visual representations of the speaker, a cross-modal fusion function f_{XM} is implemented to get the cross-modal comprehensive representations of the speaker:

$$e_t^{xm} = f_{XM}(\beta_t, p_t, A_t^{AE}) \quad (4)$$

At each time stamp t , the decoder module takes the cross-modal comprehensive representation e_t^{xm} as input and generate the listener's motion parameters β_t and p_t . The architecture of sequential decoder contains a six-layer bi-directional GRU. The decoding procedure can be formulated as:

$$\beta_{t+1}, p_{t+1} = G_{GRU}(h_t, e_t^{xm}) \quad (5)$$

where h_t is the t -th hidden state.

For the decoder model optimization, we supervise each prediction β_t, p_t with the ground truth (GT) of 3DMM coefficients in training set. The regression loss of the decoder is calculated as:

$$L_{dec} = \sum_{t=2}^T \|\beta_t - \hat{\beta}_t\|_2 + \|p_t - \hat{p}_t\|_2 + \sum_{t=2}^T w_1 \|\mu(\beta_t) - \mu(\hat{\beta}_t)\|_2 + \sum_{t=2}^T w_2 \|\mu(p_t) - \mu(\hat{p}_t)\|_2 \quad (6)$$

where $\hat{\beta}_t, \hat{p}_t$ denote the ground truth. The last two terms of Eq. 6 apply motion constraints to guarantee the inter-frame continuity, where $\mu(\cdot)$ measures the inter-frame changes.

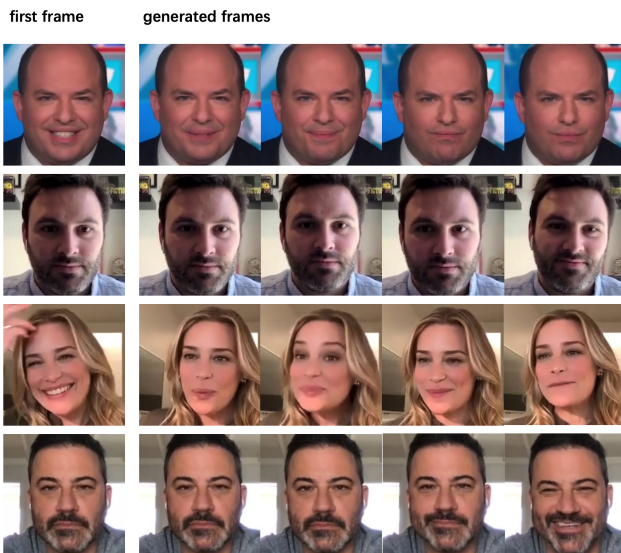
Table 2: The quantitative results of several trial submissions on evaluation system. We select several results with different components as non-rigorous ablation experiments. More rigorous ablation experiments should be re-organized.

Settings	SSIM $\uparrow$	CPBD $\uparrow$	PSNR $\uparrow$	FID $\downarrow$	CSIM $\uparrow$	PoseL1 $\downarrow$	ExpL1 $\downarrow$	PoseFD $\downarrow$	ExpFD $\downarrow$
ViCo baseline	0.495	0.152	15.335	41.319	0.538	0.120	0.197	0.015	1.497
Ours with PIRenderer	0.612	0.156	22.655	35.909	0.636	0.075	0.156	0.012	1.014
Ours with 3DFaceShop	0.639	0.161	25.563	32.290	0.639	0.072	0.141	0.010	0.997
Ours (full pipeline)	0.647	0.163	26.757	30.605	0.669	0.070	0.139	0.009	0.936

4.5 Video Renderer and Post-processing Module

The video renderer module in our pipeline refers to a controllable 3D-aware portrait generation model [14, 17], which is able to convert the predicted sequential 3DMM coefficients (expressions and head poses) to video based on a reference image. The ViCo baseline adopts PIRenderer [14] as video renderer. Through observations, it can be found that PIRenderer performs well in general, but has several defects, including background distortion and poor performance on preserving the identity when large angles. Thus we have adopted a novel state-of-the-art renderer called 3DFaceShop [17]. 3DFaceShop proposes a volume blending strategy where the renderer can form a composite output by blending dynamic and static areas, with two parts segmented from the jointly learned semantic field. With this blending strategy, 3DFaceShop could solve non-face area distortion and produce realistic portraits when viewed from free viewpoints. Experiments have demonstrated that the adoption of 3DFaceShop can lead to significant performance gains.

The final post-processing module is a video restoration method application, i.e., RVRT [8], to overcome the bottleneck due to the low-resolution and blurred training data. Specifically, we utilize pre-trained video restoration models to enhance facial details and improve visualization as much as possible.

**Figure 2: Visualization results of generated listening head video frames based on first video frame.**

5 EXPERIMENTS

5.1 Overall Evaluation on the leaderboard

As shown in Tab. 1, our proposed solution won five **1st places** on the official leaderboard out of nine evaluation metrics, and finally achieved the **1st place** in the overall score on the track of listening head generation. Our proposed solution leads the way in SSIM, PSNR, PoseL1, ExpL1 and PoseFD evaluation metrics, which indicates our generated videos are of excellent quality, both from visual perspective and from task-specified goals perspective.

5.2 Limited Ablation Study of submitted results

To verify the effectiveness of the components of our proposed solution, we selected some of the many results submitted to the online evaluation system for limited ablation experiments. Due to the large number of results submitted, the selected results do not guarantee that the experimental setups will match, and the results can only be used as a reference, with more rigorous ablation experiments to be reorganized. As shown in Tab. 2, our solution with PIRenderer has significant performance gains compared to the ViCo baseline, proving that our proposed audio encoder and sequential decoder are better able to learn the associations between speech semantics and listener head behavior. The experimental results also showed that the 3DFaceShop [17] renderer was able to alleviate some of the problems that existed in the previous PIRenderer and achieve some performance gains.

5.3 Qualitative Visualization

We show the visualization results of our solution in Fig. 2. From the reference image of the first frame in the leftmost column, one can see that our method is able to generate high-quality photorealistic video clips of the listening heads.

6 CONCLUSION

This paper is a technical report of ViCo@2023 Conversational Head Generation Challenge at the ACM Multimedia 2023 conference. Following the ViCo baseline scheme, we propose a high-performance solution by enhancing the hierarchical semantic extraction capability of the audio encoder module and optimising the decoder part, the renderer, and the post-processing modules. Our solution ranks first in the official leaderboard for the track of listening head generation.

ACKNOWLEDGMENTS

This work was supported by the Natural Science Foundation of China (NSFC62071292, U21B2013).

REFERENCES

- [1] Volker Blanz and Thomas Vetter. 1999. A morphable model for the synthesis of 3D faces. In *Proceedings of the 26th annual conference on Computer graphics and interactive techniques*. 187–194.
- [2] Zhigang Chang, Zhao Yang, Yongbiao Chen, Qin Zhou, and Shibao Zheng. 2021. Seq-Masks: Bridging the gap between appearance and gait modeling for video-based person re-identification. In *2021 International Conference on Visual Communications and Image Processing (VCIP)*. 1–5. <https://doi.org/10.1109/VCIP53242.2021.9675368>
- [3] Joon Son Chung, Jaesung Huh, Seongkyu Mun, Minjae Lee, Hee Soo Heo, Soyeon Choe, Chihyeon Ham, Sunghwan Jung, Bong-Jin Lee, and Icksang Han. 2020. In defence of metric learning for speaker recognition. *arXiv preprint arXiv:2003.11982* (2020).
- [4] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2018. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805* (2018).
- [5] Shiry Ginosar, Amir Bar, Gefen Kohavi, Caroline Chan, Andrew Owens, and Jitendra Malik. 2019. Learning individual styles of conversational gesture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3497–3506.
- [6] Jianzhu Guo, Xiangyu Zhu, Yang Yang, Fan Yang, Zhen Lei, and Stan Z Li. 2020. Towards Fast, Accurate and Stable 3D Dense Face Alignment. In *Proceedings of the European Conference on Computer Vision (ECCV)*.
- [7] Jie Hu, Li Shen, and Gang Sun. 2018. Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 7132–7141.
- [8] Jingyun Liang, Yuchen Fan, Xiaoyu Xiang, Rakesh Ranjan, Eddy Ilg, Simon Green, Jie Zhang Cao, Kai Zhang, Radu Timofte, and Luc V Gool. 2022. Recurrent video restoration transformer with guided deformable attention. *Advances in Neural Information Processing Systems* 35 (2022), 378–393.
- [9] Xian Liu, Qianyi Wu, Hang Zhou, Yinghao Xu, Rui Qian, Xinyi Lin, Xiaowei Zhou, Wayne Wu, Bo Dai, and Bolei Zhou. 2022. Learning hierarchical cross-modal association for co-speech gesture generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10462–10472.
- [10] Cheng Luo, Siyang Song, Weicheng Xie, Micol Spitale, Linlin Shen, and Hatice Gunes. 2023. ReactFace: Multiple Appropriate Facial Reaction Generation in Dyadic Interactions. *arXiv preprint arXiv:2305.15748* (2023).
- [11] Evonne Ng, Hanbyul Joo, Liwen Hu, Hao Li, Trevor Darrell, Angjoo Kanazawa, and Shiry Ginosar. 2022. Learning to listen: Modeling non-deterministic dyadic facial motion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 20395–20405.
- [12] Pascal Paysan, Reinhard Knothe, Brian Amberg, Sami Romdhani, and Thomas Vetter. 2009. A 3D face model for pose and illumination invariant face recognition. In *2009 sixth IEEE international conference on advanced video and signal based surveillance*. IEEE, 296–301.
- [13] KR Prajwal, Rudrabha Mukhopadhyay, Vinay P Namboodiri, and CV Jawahar. 2020. A lip sync expert is all you need for speech to lip generation in the wild. In *Proceedings of the 28th ACM international conference on multimedia*. 484–492.
- [14] Yurui Ren, Ge Li, Yuanqi Chen, Thomas H Li, and Shan Liu. 2021. Pirenderer: Controllable portrait image generation via semantic neural rendering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 13759–13768.
- [15] Zilong Shao, Siyang Song, Shashank Jaiswal, Linlin Shen, Michel Valstar, and Hatice Gunes. 2021. Personality recognition by modelling person-specific cognitive processes using graph representation. In *proceedings of the 29th ACM international conference on multimedia*. 357–366.
- [16] Siyang Song, Zilong Shao, Shashank Jaiswal, Linlin Shen, Michel Valstar, and Hatice Gunes. 2022. Learning person-specific cognition from facial reactions for automatic personality recognition. *IEEE Transactions on Affective Computing* (2022).
- [17] Junshu Tang, Bo Zhang, Binxin Yang, Ting Zhang, Dong Chen, Lizhuang Ma, and Fang Wen. 2022. Explicitly controllable 3d-aware portrait generation. *arXiv preprint arXiv:2209.05434* (2022).
- [18] Kaisiyuan Wang, Qianyi Wu, Linsen Song, Zhuoqian Yang, Wayne Wu, Chen Qian, Ran He, Yu Qiao, and Chen Change Loy. 2020. Mead: A large-scale audio-visual dataset for emotional talking-face generation. In *European Conference on Computer Vision*. Springer, 700–717.
- [19] Mohan Zhou, Yalong Bai, Wei Zhang, Ting Yao, and Tiejun Zhao. 2023. Interactive Conversational Head Generation. *arXiv:2307.02090* [cs.CV]
- [20] Mohan Zhou, Yalong Bai, Wei Zhang, Ting Yao, Tiejun Zhao, and Tao Mei. 2022. Responsive listening head generation: a benchmark dataset and baseline. In *European Conference on Computer Vision*. Springer, 124–142.
- [21] Mohan Zhou, Yalong Bai, Wei Zhang, Ting Yao, Tiejun Zhao, and Tao Mei. 2023. Visual-Aware Text-to-Speech. In *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 1–5. <https://doi.org/10.1109/ICASSP49357.2023.10095084>
- [22] Hao Zhu, Man-Di Luo, Rui Wang, Ai-Hua Zheng, and Ran He. 2021. Deep audio-visual learning: A survey. *International Journal of Automation and Computing* 18 (2021), 351–376.